

JAYENDRA PAWAR

Software Engineer — Go | Distributed Systems | Kubernetes | Cloud Infrastructure

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WORK EXPERIENCE

IBM

Software Engineer II (Platform/Backend)

Oct 2024 – Present

Bengaluru, Karnataka

Distributed Worker Orchestration

- Architected a highly available, **Go-based Kubernetes controller** for **10K+ workers**, automating worker lifecycle management (provisioning, replacement & decommissioning) and **reducing provisioning latency by 30%**
- Orchestrated **safety-gated cluster automation** for maintenance & node upgrades, automating unhealthy-node replacement and unclaimed-node cleanup while **improving cluster utilization by 25%** and **reducing recurring on-call escalations**
- Designed a **zone-aware auto-scaling strategy** balancing resource utilization vs. availability across **multi-AZ clusters**, sustaining **99.9% uptime** while **eliminating over-provisioning**
- Developed an **integration-test framework** on dedicated Kubernetes clusters, validating **platform controllers** for worker orchestration, reconciliation, scaling, resilience & recovery across **12+ managed data services** as a **production-readiness gate**

Instance Networking & Access Control

- Led the **end-to-end development** of **instance network controls in Go**, covering REST API design, event-driven network updates, technical specifications and infrastructure planning; **delivered SDK/CLI support for enterprise customers**
- Engineered **instance network-state synchronization** using Kubernetes informers, rate-limited queues and batch processing, enabling deduplication, idempotency, retries, and failure isolation to meet a **10sec network-state consistency SLA**
- Implemented **instance-level network policy enforcement**, evaluating user-defined rules on connection requests to authorize **instance-to-instance connectivity**, replacing broad service-wide permissions with **fine-grained access controls**

Cloud Infrastructure & Reliability

- Led **compute migration & resource optimization**, reducing infrastructure **costs by ~14% (\$166K)** while mitigating outage risks
- Built a **Slack-based ChatOps** platform providing **30+ Kubernetes & cloud-service** operations, **reducing operational toil** and accelerating incident resolution
- Standardized **Deployable Architecture (IaC) modules**, enabling **40% faster feature rollouts** and simplifying **IAM configuration** across teams while strengthening system resilience
- Established **CI/CD pipelines** using **GitHub Actions**, with pre-merge validation, security scanning, and quality gates, enabling **zero-downtime deployments** across **on-premises Kubernetes clusters**

Freshworks

Software Engineer (Backend)

Jul 2022 – Sept 2024

Bengaluru, Karnataka

Payment Services & Integrations

- Engineered a **highly available distributed payment service** integrating **Stripe APIs** in **FreshPay**, enabling critical **payment workflows** such as bind flow, authorization, tokenization, payments, and refunds; serving **4.9 million users** in Europe
- Developed a **scalable asynchronous notification service**, incorporating **API specifications, ERD designs**, and workflows using **gRPC** and **schedulers**, supporting **150K+ customers**
- Led the **migration of payment services to EKS (Kubernetes)**, achieving a **30% cost reduction** while enhancing **deployment efficiency** and **scalability** for millions of daily transactions
- Accelerated **MySQL performance**, cutting **query execution time by 60%** through **indexing, partitioning**, and **execution plan optimization**. Implemented **CDC-based caching** to synchronize dynamic and static data in real time, **reducing API response time by 40%**
- Deployed **canary releases** across **5 microservices**, minimizing deployment risk and **reducing rollback rate by 15%** while ensuring adherence to release policies, performance benchmarks, and SLOs

EDUCATION

National Institute of Technology, Jaipur

Bachelor of Technology in Mechanical Engineering; Minor in Computer Science

Jul 2018 – Jun 2022

Jaipur, Rajasthan

- Awarded **Aditya Birla Scholarship** — ranked among the **top 15 candidates across all NITs (2021)**

SKILLS

- Programming Languages:** Go, Python, Rust, C/C++
- Backend & Distributed Systems:** REST APIs, gRPC, WebSockets, Kafka, RabbitMQ, Concurrency, Event-Driven Architectures
- Cloud & Infrastructure:** Kubernetes, Docker, Linux, Helm, Terraform, GitHub Actions, Jenkins, Reliability Engineering
- Databases & Observability:** MySQL (Indexing, Partitioning), Redis, Prometheus, Grafana, Monitoring & Alerting
- Core Areas:** Data Structures & Algorithms, Object-Oriented Design, System Design, Microservices, Performance Optimization